<WA/>
2026

Forms

The Foundations of User Interaction

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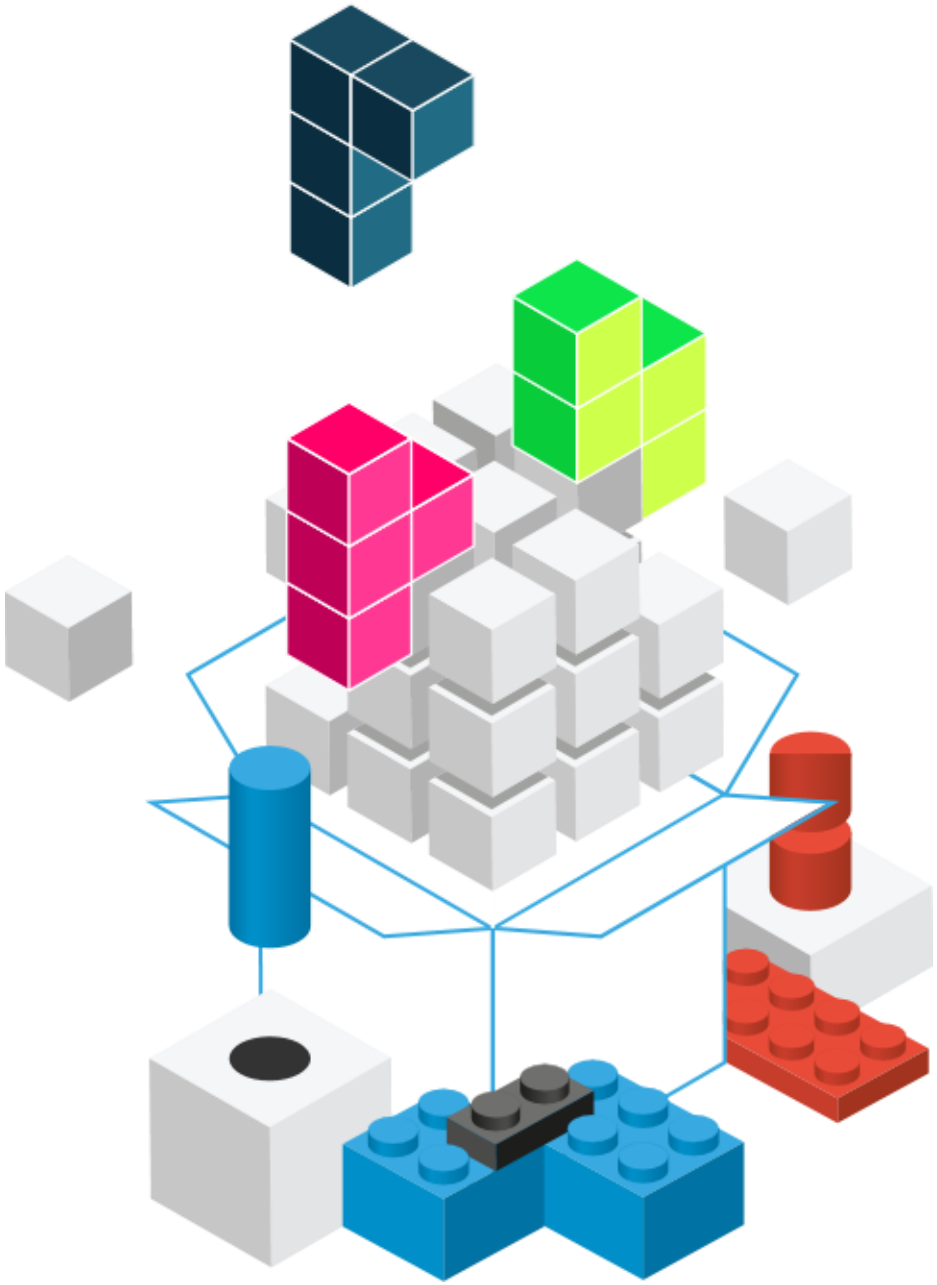

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https://react.dev/reference/react-dom/components#form-components

Full Stack React, Chapter “Forms”

React Handbook, Chapter “JSX”

Forms, Events and Event Handlers

FORMS IN JSX

HTML Forms

- (Native) HTML Forms are *inconsistent*: different ways of handling values, events etc. depending on the type of input element
  - Consequence of backward compatibility
- For instance:
  - `onChange` on a radio button is not easy to handle
  - `value` in a `textarea` does not work, etc.
- React flattens this behavior exposing (via JSX) a more uniform interface
  - Synthetic Events

Value in JSX forms

- The **`value`** attribute always holds the current value of the field
- The `defaultValue` attribute holds the default value that was set when the field was created
- Other examples:
  - `textarea`: the content is in the `value` attribute; it is NOT to be taken from the actual content of the `<textarea>…</textarea>` tag
  - `select`: do not use the `<option selected>` syntax, but `<select value='id'>`

Change Events in JSX Forms

- React provides a more consistent **onChange** event
- By passing a function to the `onChange` attribute you can subscribe to events on form fields (every time `value` changes)
- `onChange` fires when typing a <u>single</u> character into an `input` or `textarea` field
- It works consistently across fields: even `radio`, `select` and `checkbox` input fields fire a `onChange` event

Event Handlers

- An Event Handler callback function is called with one parameter: an event object
- All event objects have a standard set of properties
  - `event.target`: *source* of the event
- Some events, depending on categories, have more specific properties

Synthetic Events

https://react.dev/reference/react-dom/components/common#react-event-object

- "High level events" wrap the corresponding DOM Events
- Same attributes as `DOMEvent`
- `target` points to the source of the event.
- In case of a *form element*
  - `target.value` = current input value
  - `target.name` = input element name

```
boolean bubbles
boolean cancelable
DOMEventTarget currentTarget
boolean defaultPrevented
number eventPhase
boolean isTrusted
DOMEvent nativeEvent
void preventDefault()
boolean isDefaultPrevented()
void stopPropagation()
boolean isPropagationStopped()
DOMEventTarget target
number timeStamp
string type
```

Synthetic Events

https://reactjs.org/docs/events.html

Category	Events
Clipboard	onCopy onCut onPaste
Composition	onCompositionEnd onCompositionStart onCompositionUpdate
Keyboard	onKeyDown **onKeyPress** onKeyUp
Focus	**onFocus onBlur**
Form	**onChange** onInput **onInvalid** onReset **onSubmit**
Generic	onError **onLoad**
Mouse	**onClick** onContextMenu onDoubleClick onDrag onDragEnd onDragEnter onDragExit onDragLeave onDragOver onDragStart onDrop onMouseDown onMouseEnter onMouseLeave onMouseMove onMouseOut onMouseOver onMouseUp
Pointer	onPointerDown onPointerMove onPointerUp onPointerCancel onGotPointerCapture onLostPointerCapture onPointerEnter onPointerLeave onPointerOver onPointerOut
Selection	onSelect
Touch	onTouchCancel onTouchEnd onTouchMove onTouchStart
UI	onScroll
Wheel	onWheel
Media	onAbort onCanPlay onCanPlayThrough onDurationChange onEmptied onEncrypted onEnded onError onLoadedData onLoadedMetadata onLoadStart onPause onPlay onPlaying onProgress onRateChange onSeeked onSeeking onStalled onSuspend onTimeUpdate onVolumeChange onWaiting
Image	onLoad onError
Animation	onAnimationStart onAnimationEnd onAnimationIteration
Transition	onTransitionEnd

Tip: Defining Event Handlers

- Define the function as…
  - an arrow function
  - a function expression

```
const handler = () => { ... }
```

```
handler = function() { ... }
```

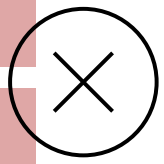
Tip: Defining Event Handlers

- Pass the *name* of the function as a prop
  - As a function object (not string)
  - Do NOT *call* the function

```
return <div handler={handler} />
```

```
return <div handler={handler()} />
```

```
return <div handler='handler' />
```



Tip: Defining Event Handlers

- Specify the *name* of the function prop in the event handler

- If you need to pass *parameters*, use an *arrow* function

```
return <button onClick=
  {props.handler} />
```

```
return <button onClick=
  {props.handler()} />
```

```
return <button onClick=
  {props.handler(a, b)} />
```

```
return <button onClick=
  {()=>props.handler()} />
```

```
return <button onClick=
  {()=>props.handler(a, b)} />
```

Who Owns The State?

- Form elements are inherently stateful: they hold a value
  - Input text form, selection, etc.
- But: React components are the ones designed to handle the state
- The props and state are used to render the component
  - To correctly render the component from the virtual DOM, React needs to know which value must be set in the form element
  - Hence, on every change (onChange) React *must be notified* to get the new value and update the component state

Where Is The Source of Truth?

Controlled Form Components

- When the React component holds, in its state, the value to be shown in the form element, it is named a **controlled** form component

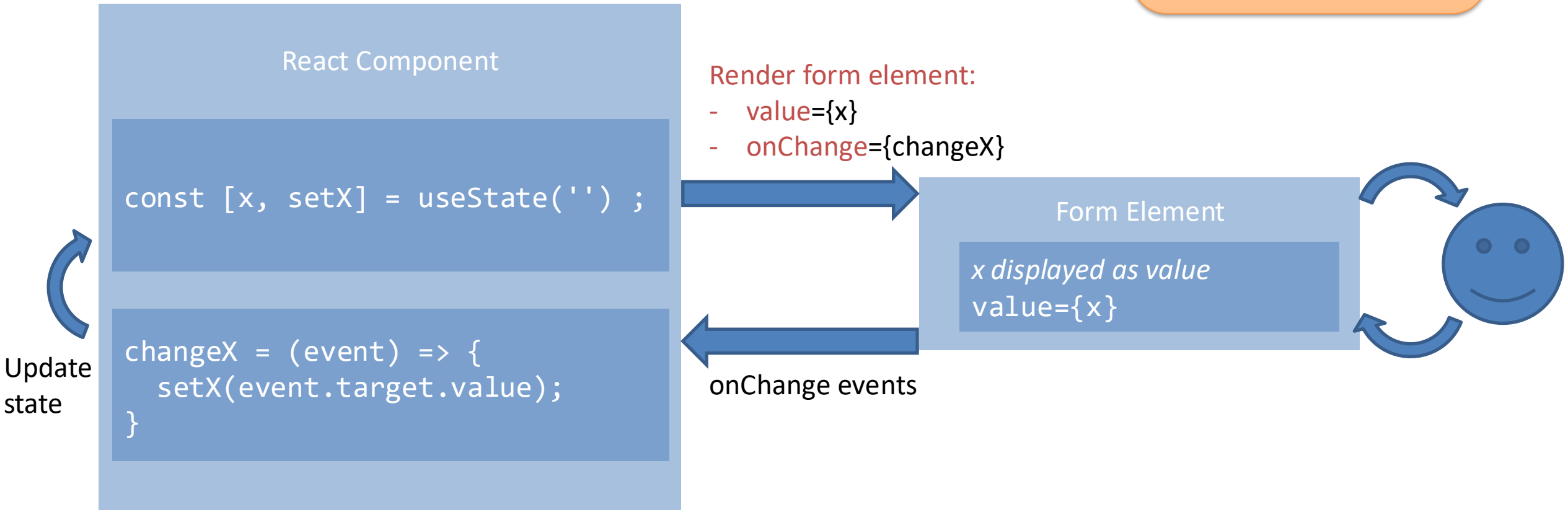
Preferred!

Uncontrolled Form Components

- In some occasions, it could be useful to keep the value directly in the HTML form element in the DOM: **uncontrolled** form component
- **NOT** recommended in this course

Controlled Form Components

Setting `value` + onChange makes the form component fully controlled



Controlled Form Component

- The event handler changes the state, *setXXX()* starts the update of the virtual DOM that then updates the actual DOM content

```
function MyForm (props) {
  const [name, setName] = useState('');
  return <form onSubmit={handleSubmit}>
    <label> Name:
      <input type="text" value={name}
         onChange={handleChange} />
    </label>
    <input type="submit" value="Submit" />
  </form> ;
}
```

```
const handleSubmit = (event) => {
  console.log('Name submitted: ' +
    name);
  event.preventDefault();
}

const handleChange = (event) => {
  setName(event.target.value) ;
};
```

Uncontrolled Form Components

Not setting `value` + onChange makes the form component uncontrolled

React Component

NO React state

```
submitForm = (event) =>
{
   props.saveData(…);
}
```

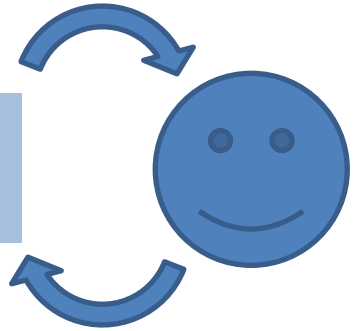
Render Form Element
- defaultValue={props.x}
- onSubmit={submitForm}

Form Element

x displayed as initial value

onChange

onSubmit



Uncontrolled components will NOT be described NOR used

Tip: Form Submission

- The `onSubmit` event is generated by the `<form>` element
- Always call `event.preventDefault()` to avoid the submission (and reloading of the page)
- Perform *validation* of *all form data* before proceeding
  - Using checks on `state` variables (on a controlled component, they contain updated information)
  - May use validator https://github.com/validatorjs/validator.js

Alternatives to Controlled Form Components

- Sometimes, it is tedious to use controlled form components
  - Need to write an event handler for every way data can change
  - Pipe all of the input state through a React component
- Alternatively, use a **library** such as Formik
  - Keep things organized without hiding them too much
  - Form state is inherently ephemeral and local: does not use state management solutions (e.g., Redux/Flux) which would unnecessary complicate things
  - Includes validation, keeping track of the visited fields, and handling form submission
  - https://jaredpalmer.com/formik

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